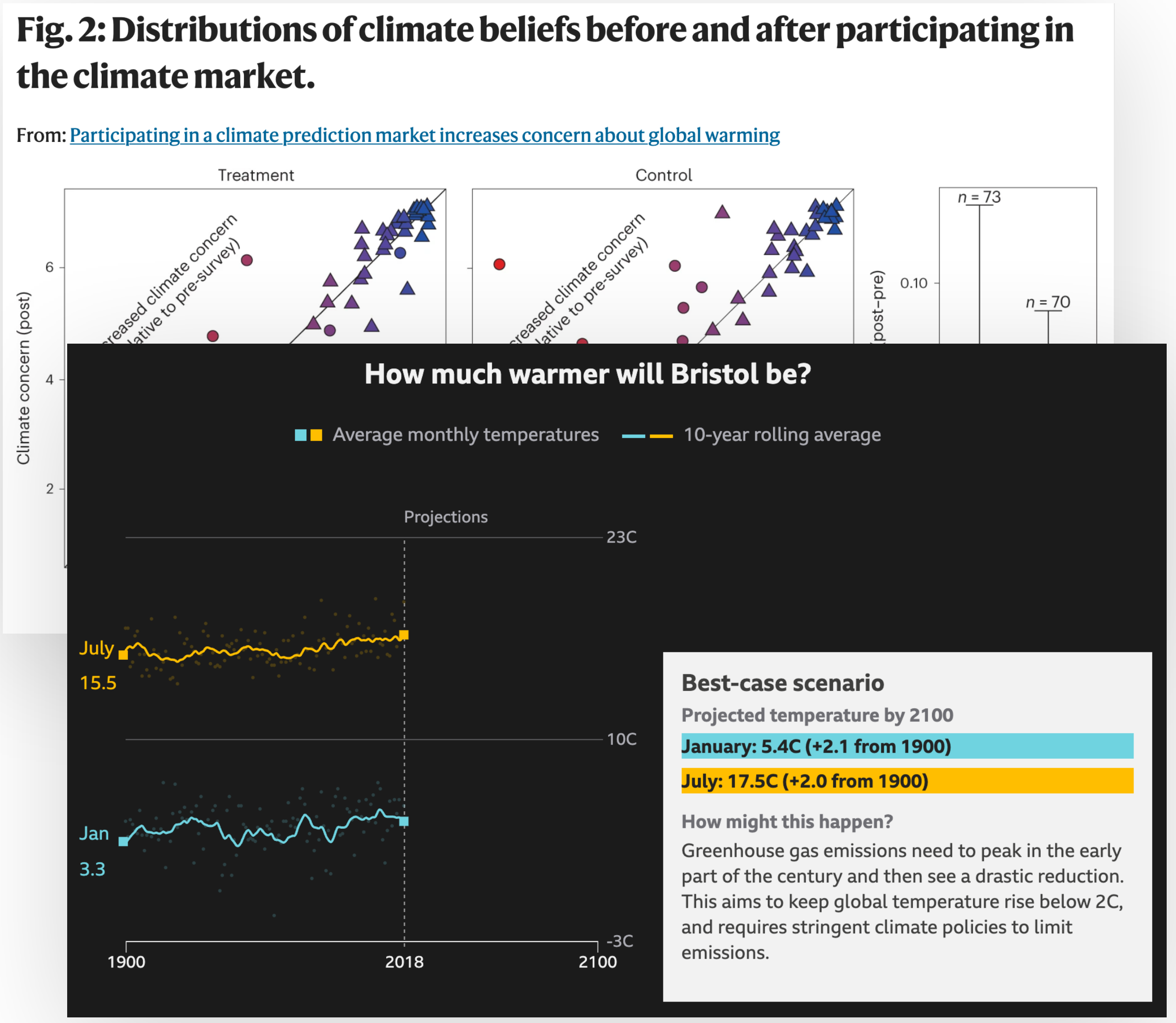
Fluid: A Transparent Programming Language

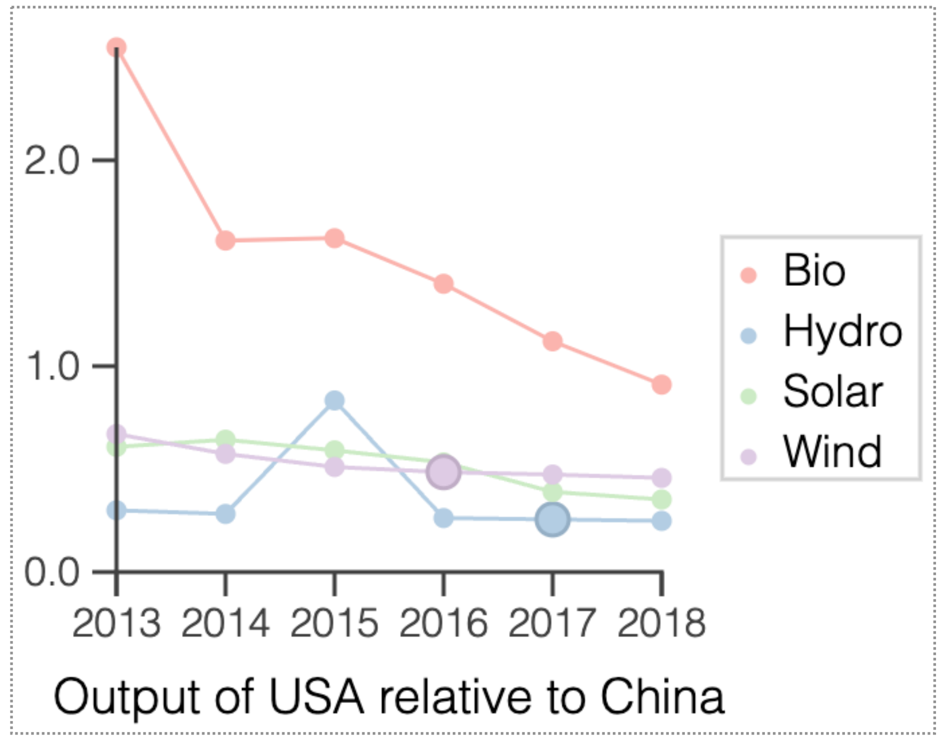
Research prototype at http://f.luid.org

Problem



Research papers and news articles are opaque — hard to critique, understand or trust

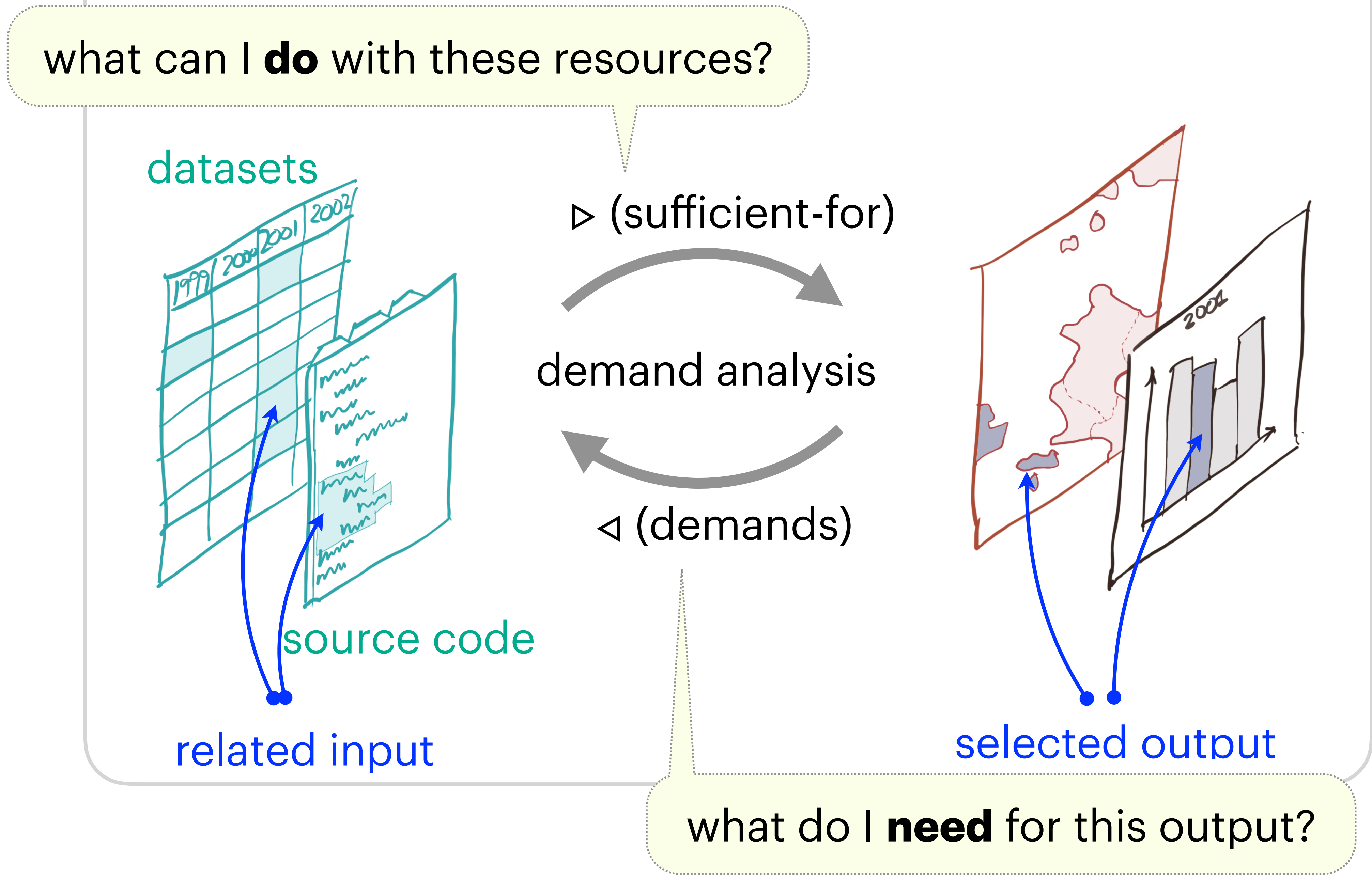
Research goal



year	*energyType*	*China*	*USA*	*=USA/China*
2016	Wind	169	82	0.49
2017	Hydro	313	80	0.26

Data-driven artefacts able to reveal relationship to underlying data

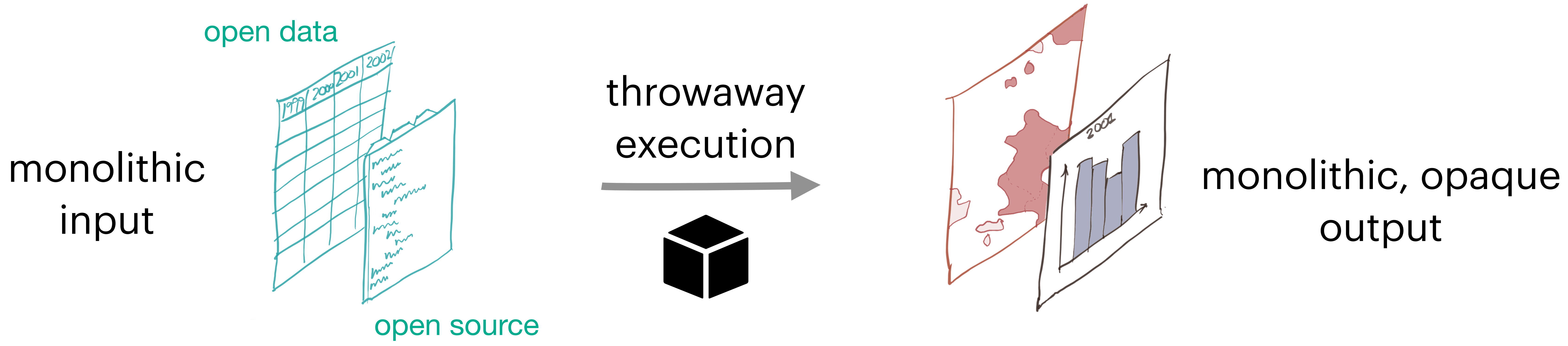
Methodology



Adjoint operators ◁ ⊣ ▷ and De Morgan duals ▶ ⊣ ◀ exposing fine-grained I/O relationships

Candidates for inclusion

Automated, correct-by-construction transparency
Enrich with intensional information?



Crucial **relationships** between data and outputs established during execution are lost and are very hard to reconstruct after the fact

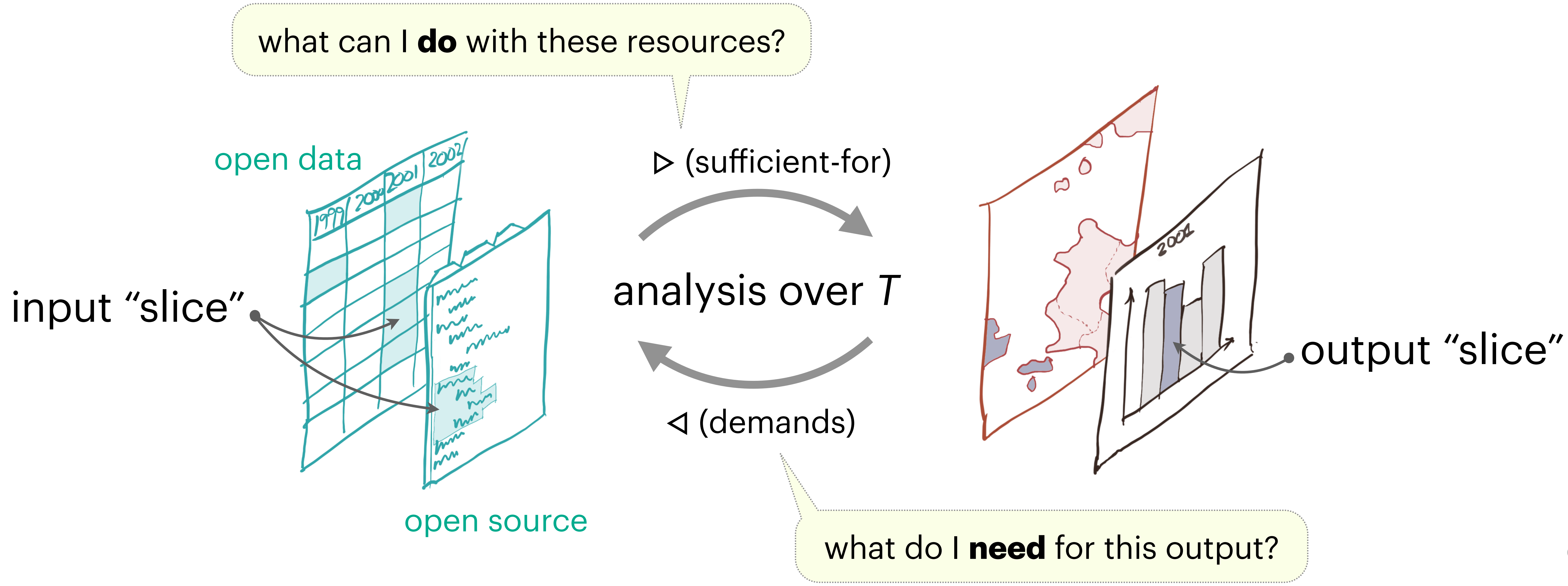
My research: can techniques from **programming languages** help us rethink this picture?

Fluid: A Transparent Programming Language

Replace run-once execution by **persistent and fine-grained bidirectional relationship**

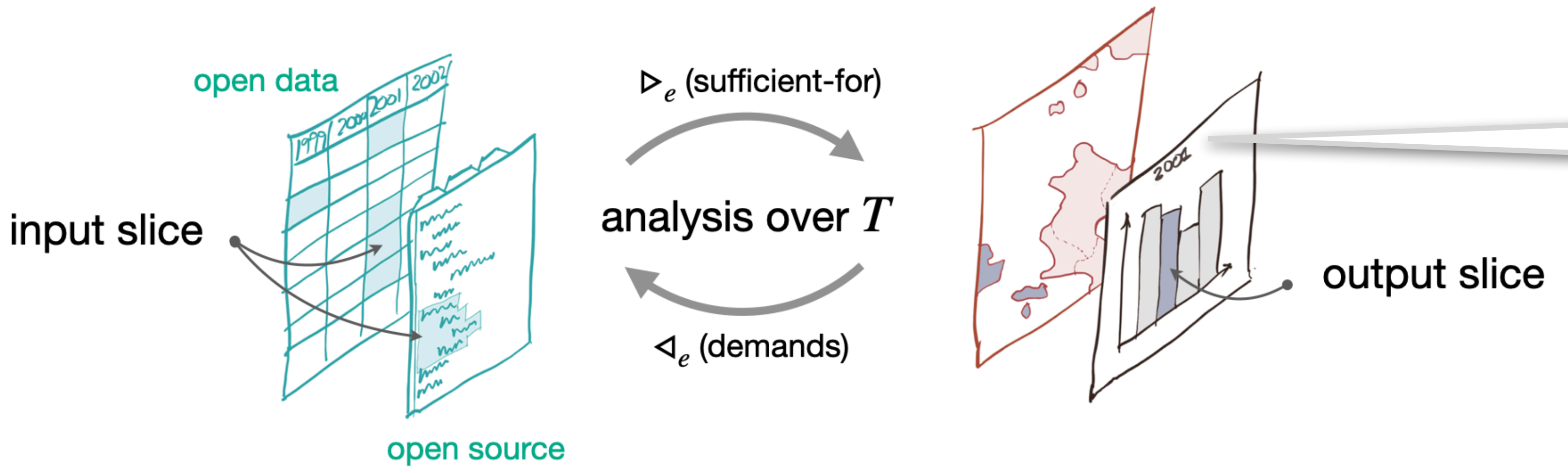
program produces a **trace** T as well as an output

pair of functions $\triangleleft \dashv \triangleright$ relating input and output “slices” called a **Galois connection**



Bake this metadata into computed artefact as a kind of **built-in transparency**

Linked outputs (i)

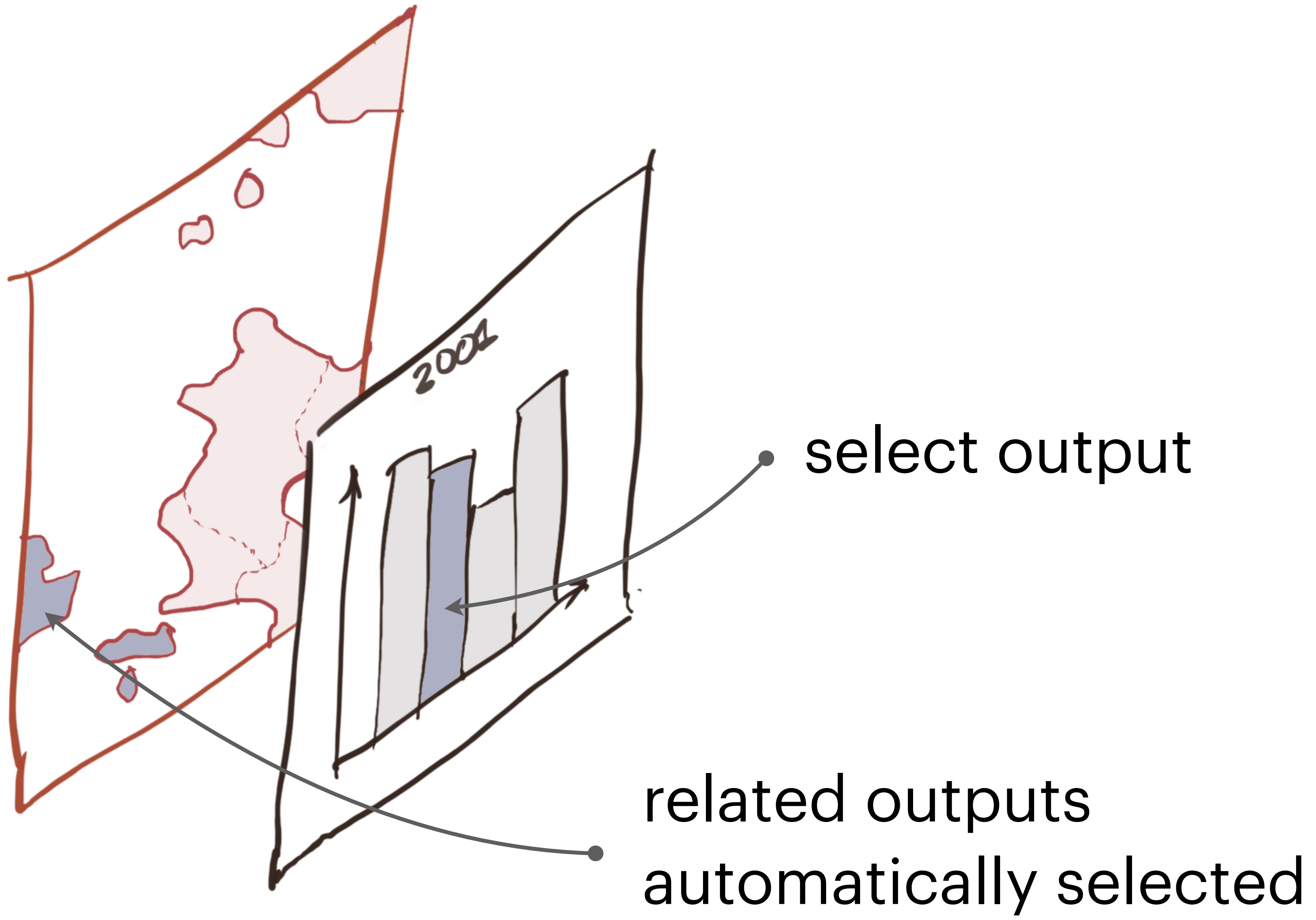


What else might support intelligibility?
What about how **outputs relate to other outputs**?

Our insight:
Outputs *x* and *y* are related if they **compete for inputs** i.e. $\triangleleft x$ and $\triangleleft y$ overlap

however: not easy to compute related outputs using only $\triangleleft$!

brushing and linking



In data visualisation, **brushing and linking** is an important comprehension tool, but ad hoc and poorly understood

Linked outputs (ii)

Showed how to define **related-outputs** operator ◁▶
that picks out “competing outputs” by allowing slices to have **complements**

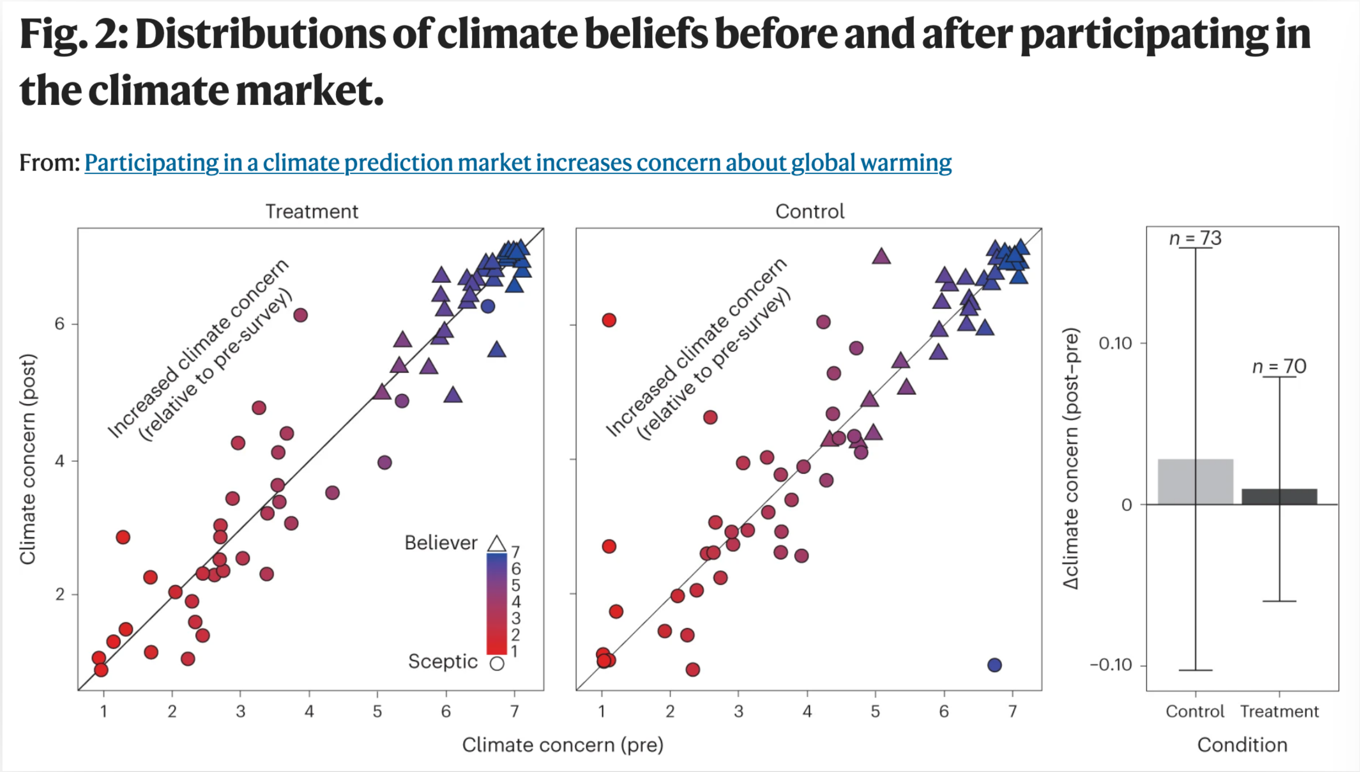
Transparent research artefacts (i)

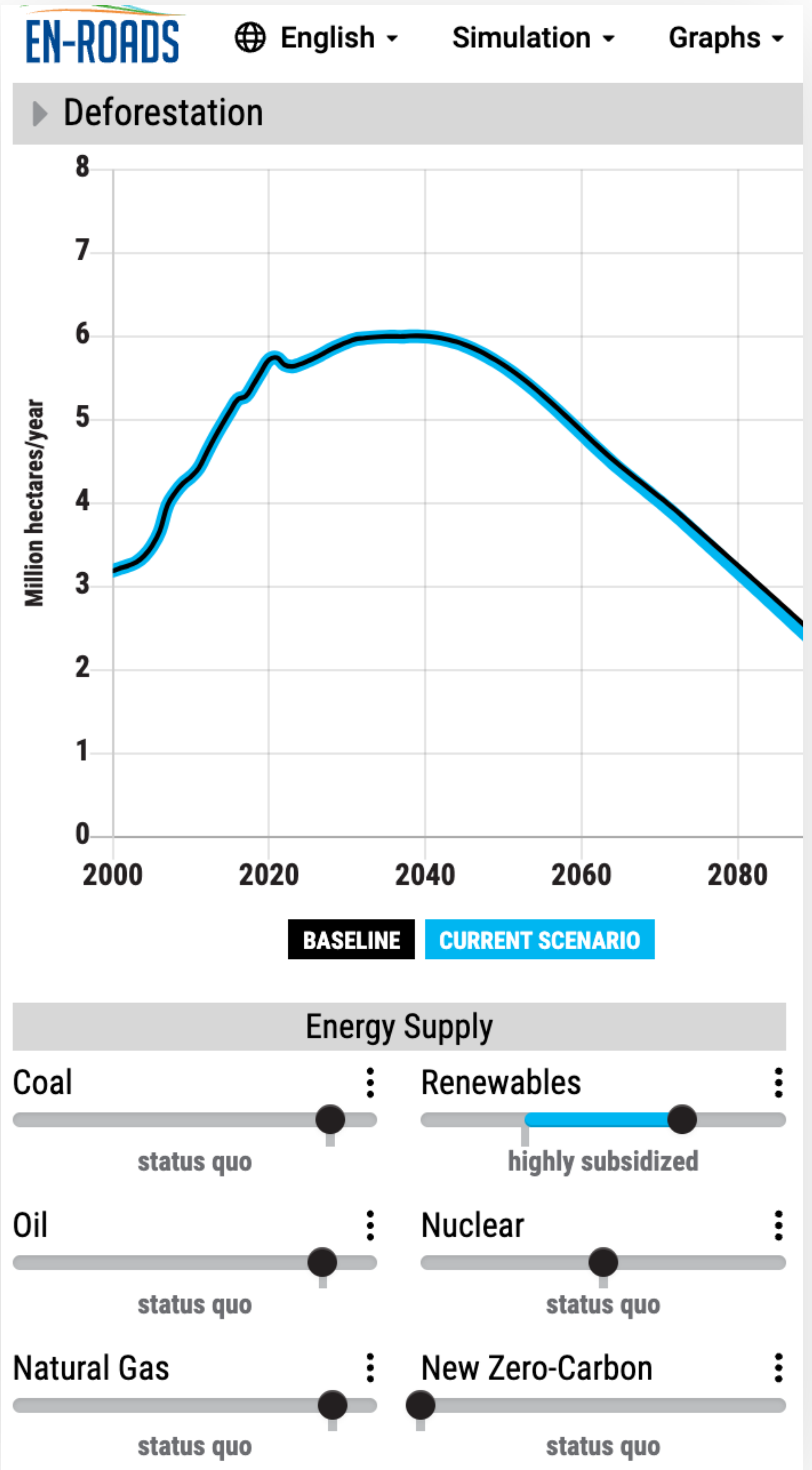
Today: papers, news articles and educational tools are opaque

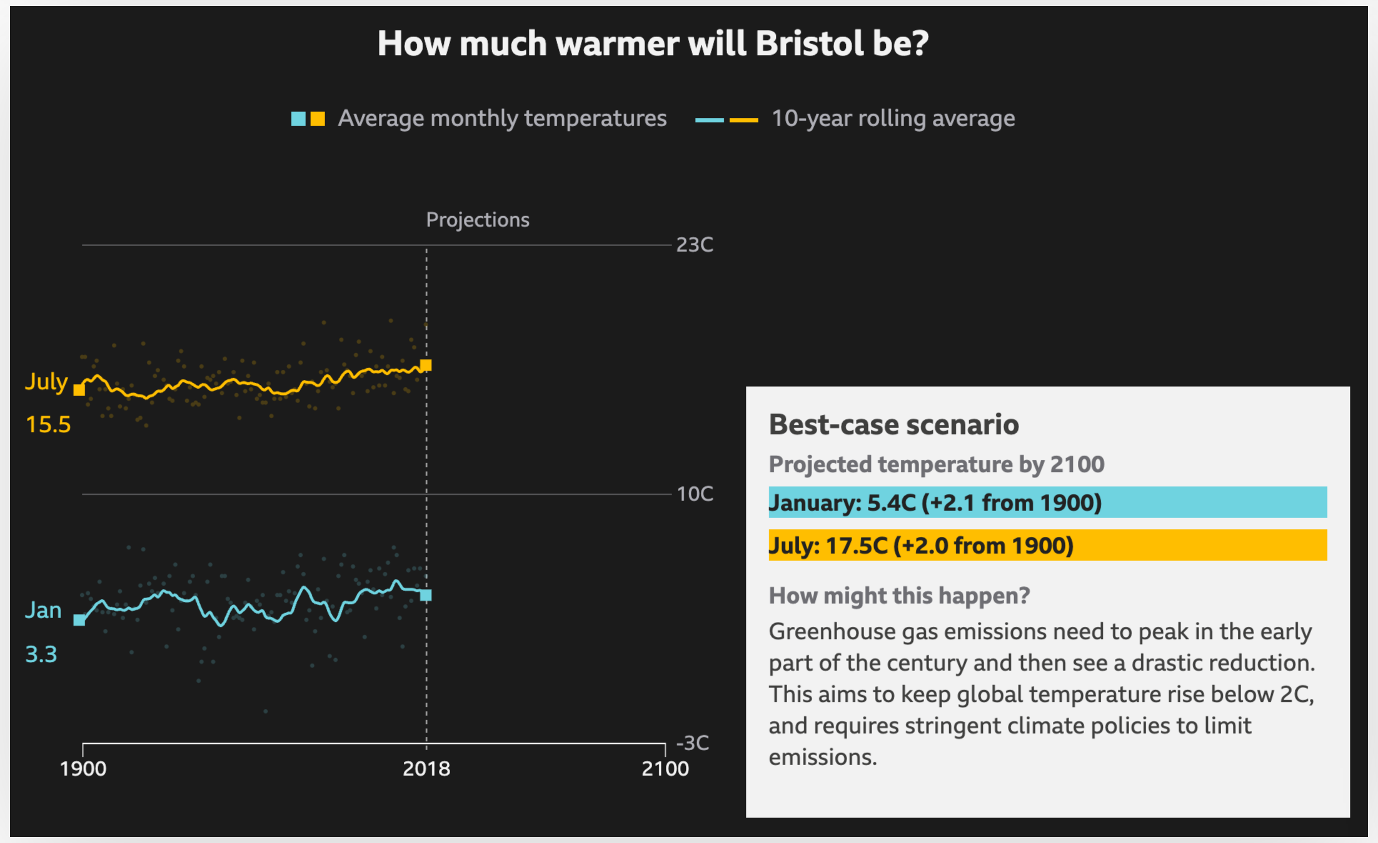
Goal: online artefacts which are **transparent and self-explaining** via rich automated interactions, building on Fluid prototype

Climate science communication today

research papers





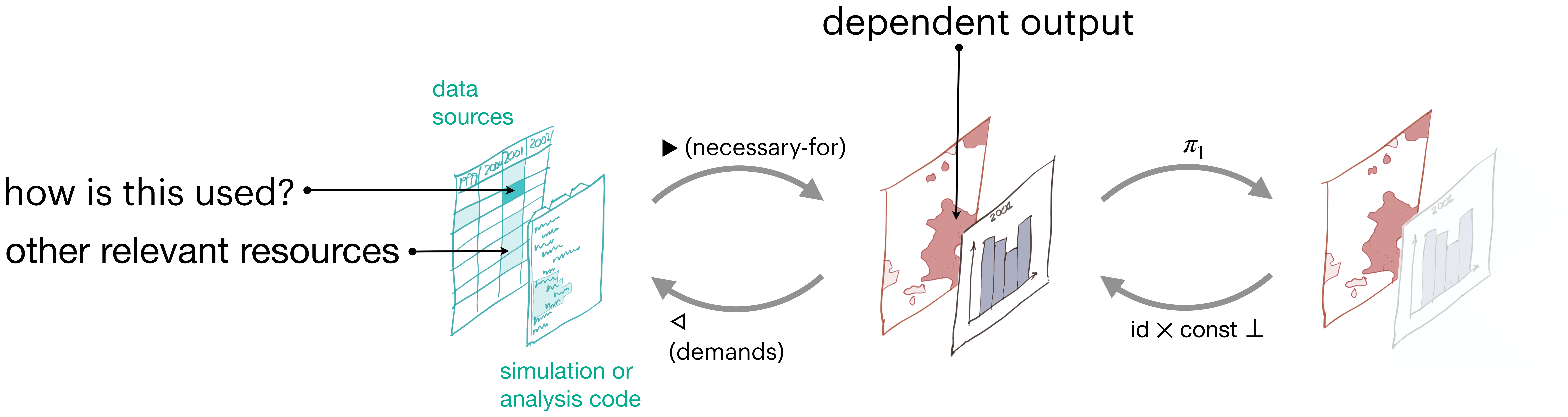


visual journalism

interactive simulation

Transparent research artefacts (ii)

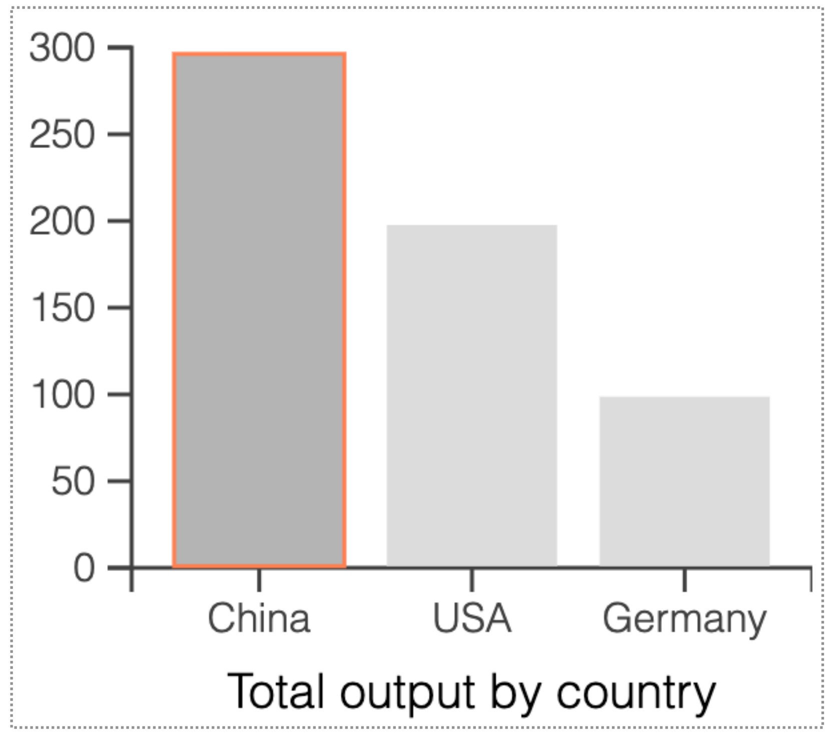
Automated interactions to support **comprehension** and **reuse**



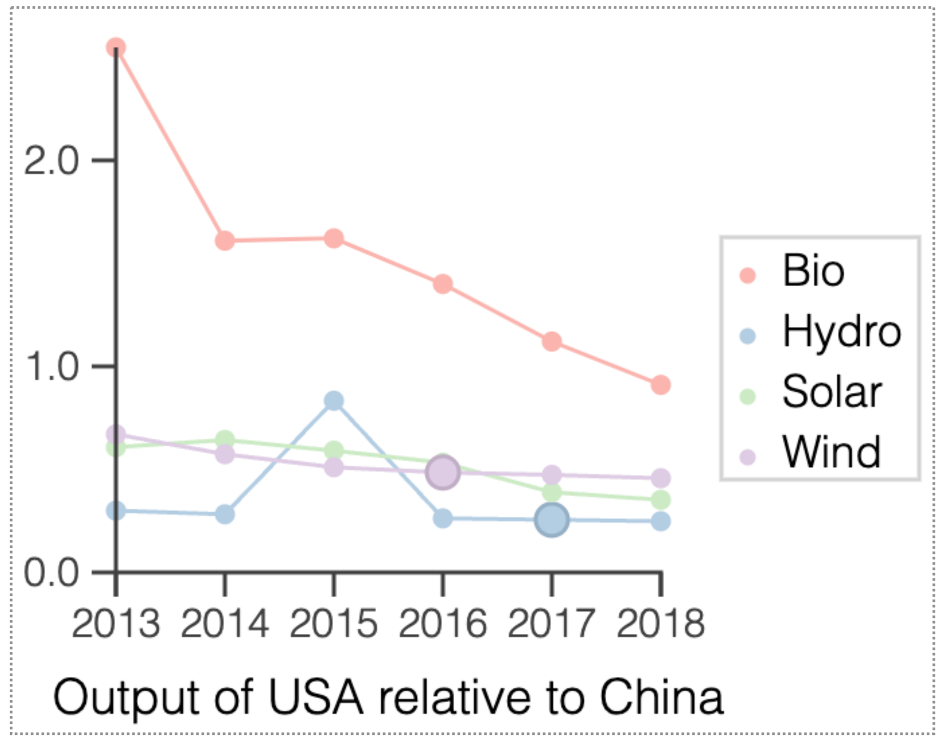
related-inputs := ▶◁ (demands ∘ necessary-for)

Other adjoint operators for focusing views on **geographical regions** or **time slices**

Self-explaining computation: equip output selections with intensional explanations (how, not just what)



energyType	***output***
Bio	10.3
Hydro	96
Solar	44
Wind	14.5
Sum	$\boxed{\textbf{164.8}}$



year	***energyType***	***China***	***USA***	***=USA/China***
2016	Wind	169	82	$\boxed{0.49}$
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Use to implement climate articles and explorable simulations like En-ROADS that are **transparent and self-explanatory**